S6 ADDITIONAL RESULTS

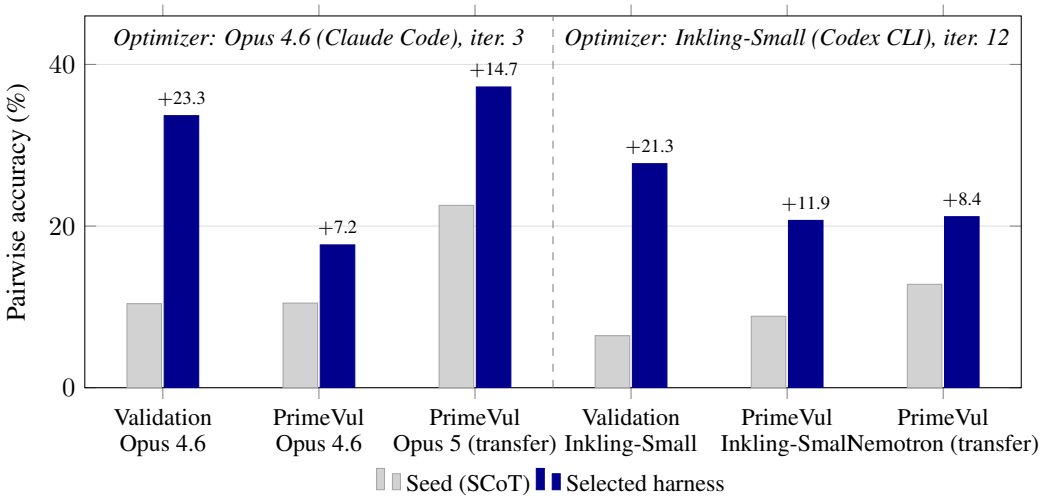


Figure S6.1: Pairwise accuracy of the seed and the harness selected on validation; labels give the gain in points. Transfer bars use a model that took no part in optimization.

S6.1 EXPANDED PER-ITERATION PLOTS

Figs. S6.2–S6.5 repeat the columns of Fig. 3 at full width, with the value of every point printed. Light yellow bands mark the Pareto frontier on validation and stars mark the best pairwise accuracy of each series, as in Fig. 3.

Due to constrained hosted-model inference token budgets, we ran four refinement iterations with Opus 4.6 and five with GPT-5.4-mini rather than the 15 iterations used for Inkling-Small.

GPT-5.4-mini improved pairwise validation accuracy from 16.83% for the seed harness to 23.27% at iteration 1, a gain of 6.44 points. The validation-selected harness achieved 16.78% on PrimeVul. Because a matching held-out seed evaluation wasn't recorded, we report the absolute held-out score rather than an improvement.

Table S6.1: Reconstructed Inkling-Small token usage per validation sample.

Iter	harness	Completion	Total
0 (seed)	915.9	605.0	1,520.9
1	1,297.9	654.7	1,952.6
2	1,515.9	665.6	2,181.5
3	1,822.4	602.8	2,425.3
4	3,823.3	825.7	4,649.0
5	1,845.7	696.1	2,541.9
6	1,791.1	738.3	2,529.4
7	1,930.1	761.7	2,691.8
8	1,758.1	938.6	2,696.7
9	1,930.1	929.0	2,859.1
10	1,933.1	896.4	2,829.5
11	2,010.1	840.4	2,850.6
12 (selected)	2,105.1	809.2	2,914.4
13	2,387.1	808.0	3,195.1
14	2,295.1	787.5	3,082.6
15	2,786.1	915.2	3,701.3

Means over 404 functions, using the official tokenizer on logged text. Excludes hidden Codex instructions and discarded retries.

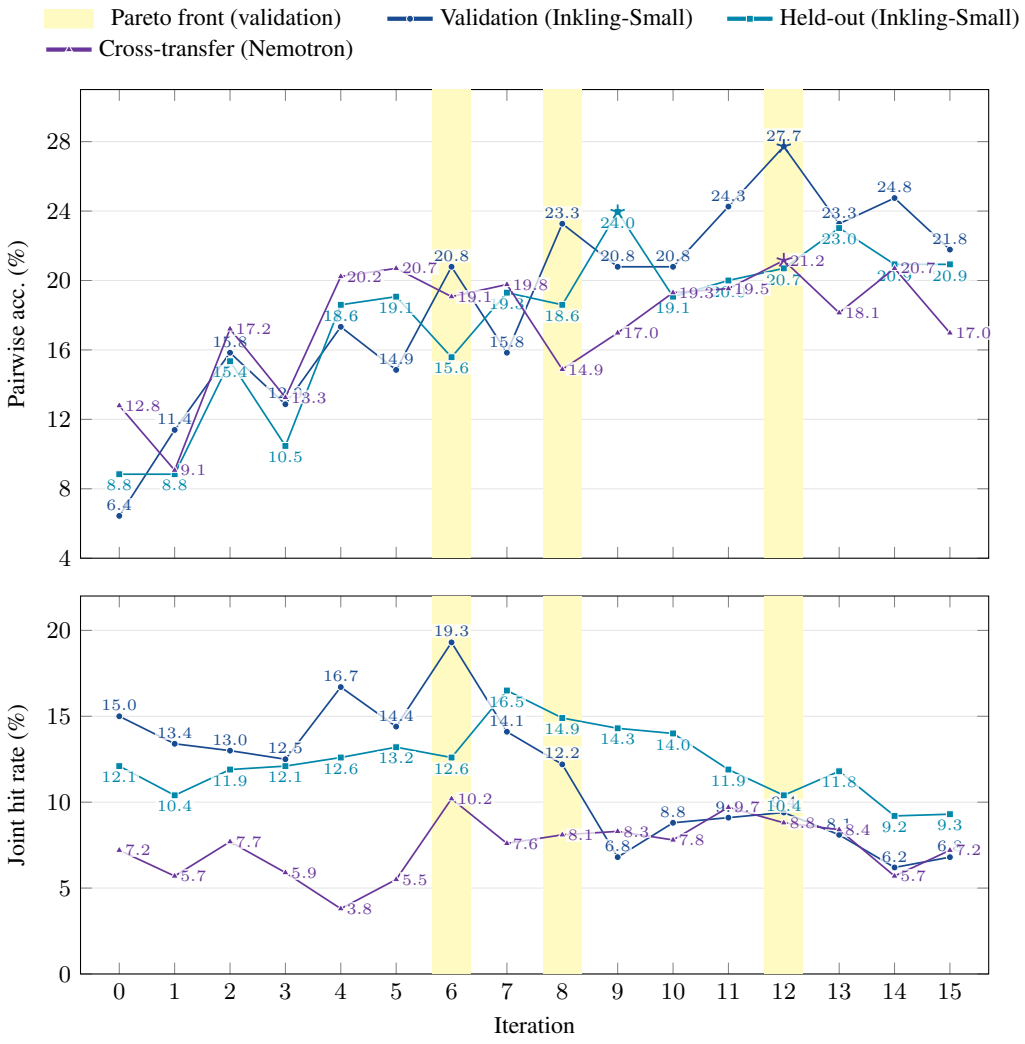


Figure S6.2: Expanded Fig. 3(a): Codex + Inkling-Small.

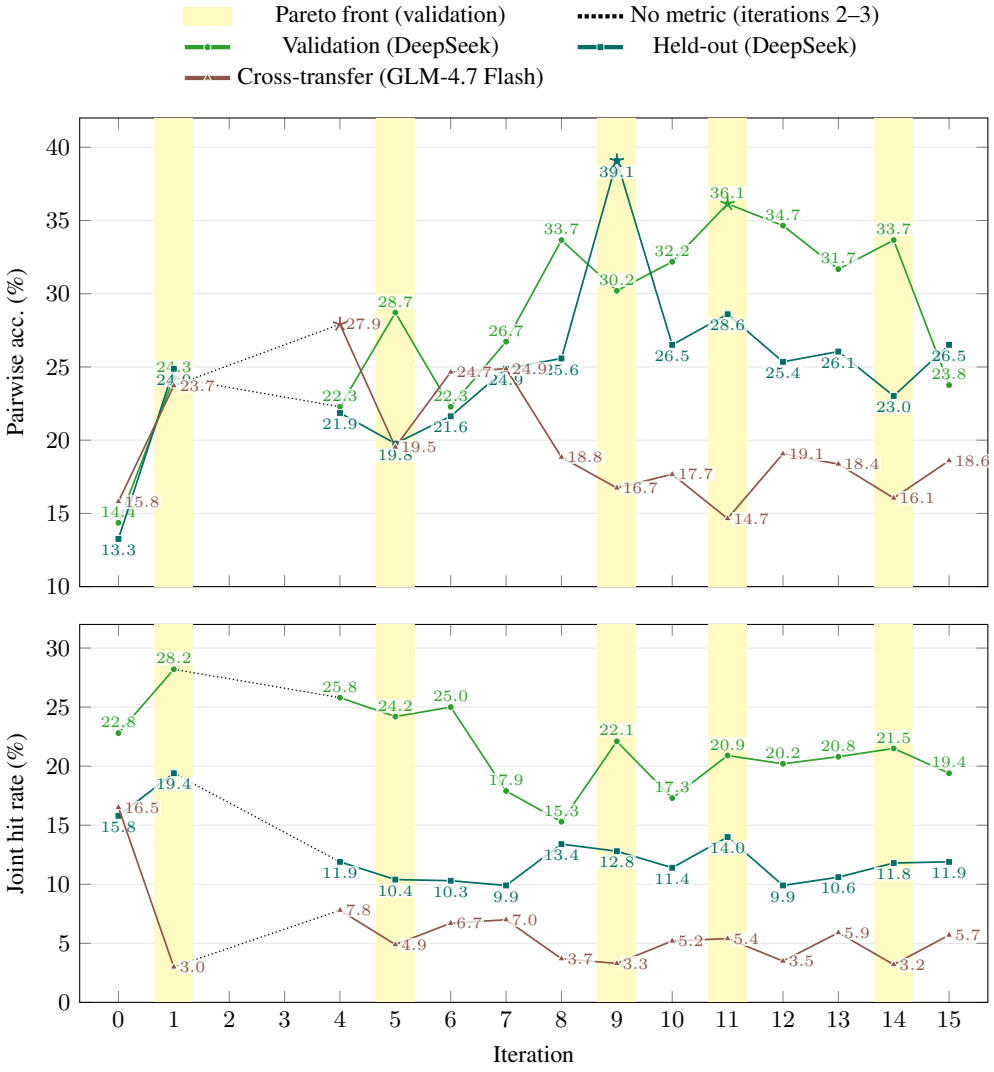


Figure S6.3: Expanded Fig. 3(b): Codex + DeepSeek V4 Flash.

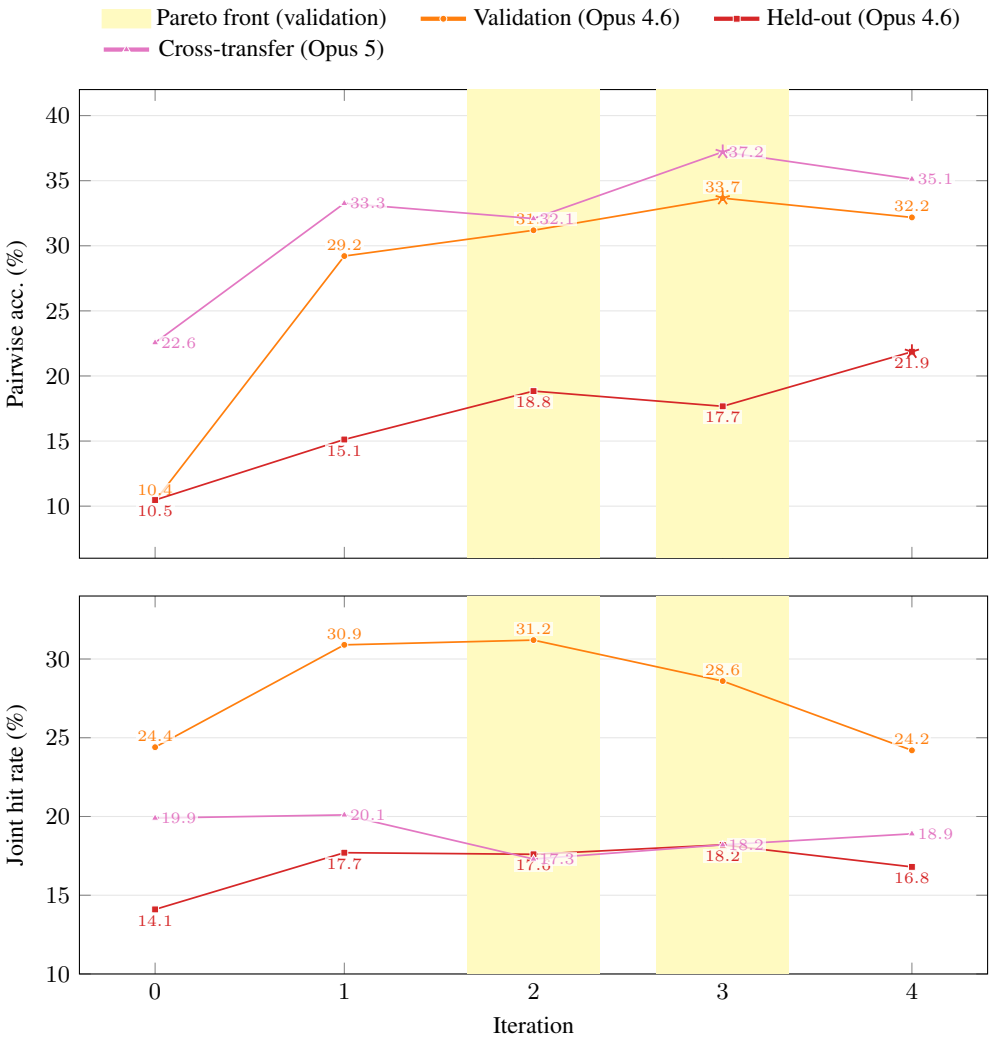


Figure S6.4: Expanded Fig. 3(c): Claude Code + Opus 4.6.

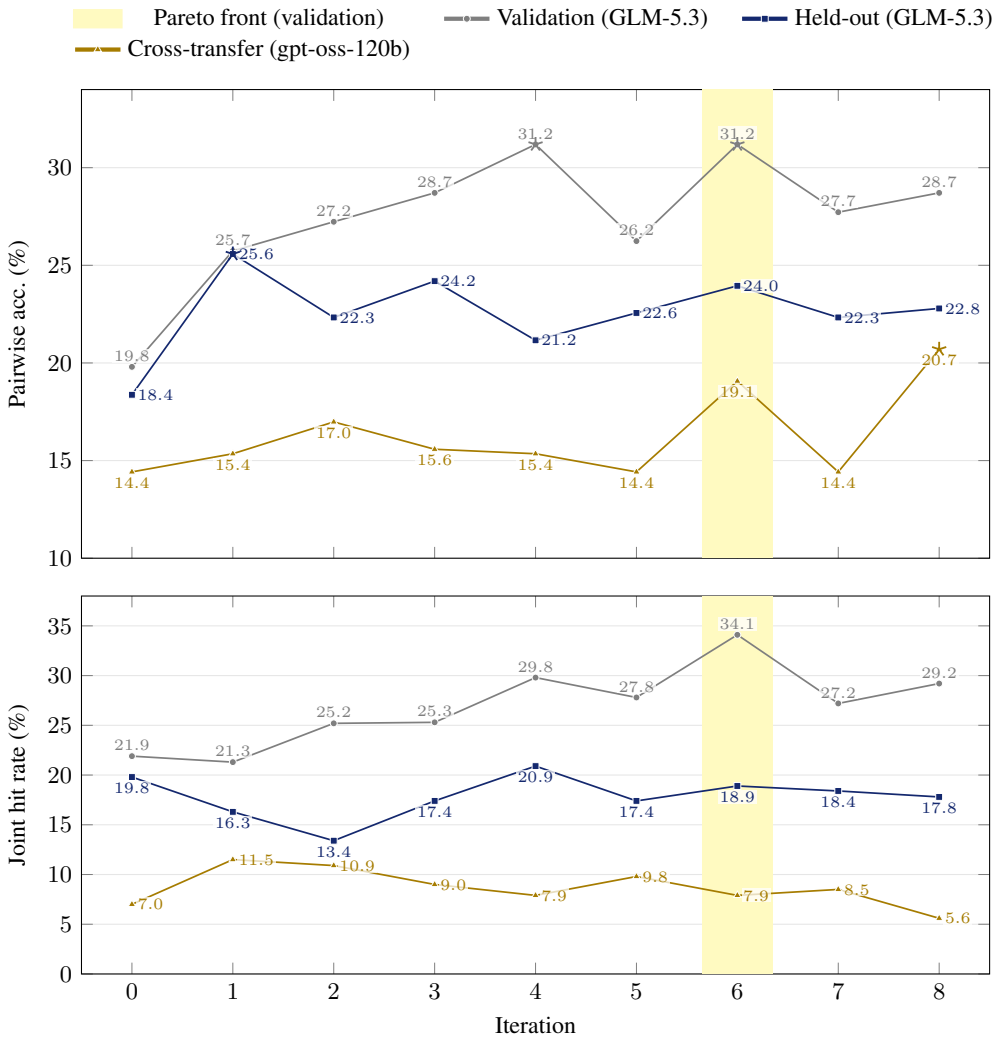


Figure S6.5: Expanded Fig. 3(d): Claude Code + GLM-5.3.

S6.2 WORKED EXAMPLE: SELECTING THE PARETO FRONTIER

We use the Inkling-Small run (Fig. S6.2) to show how MetaVul selects its Pareto frontier. Each of the 16 evaluated harnesses (the seed and iterations 1–15) is a point with two validation objectives, pairwise accuracy and joint hit rate, both maximized (Table S6.2). We say harness A *dominates* harness B if A scores no lower than B on both pairwise accuracy and joint hit rate, and higher than B on at least one of those metrics. The frontier contains those harnesses which are not dominated by any other harness. If two or more harnesses have identical scores on both metrics, we keep only the earliest one.

- Iteration 12 has the highest pairwise accuracy (27.7), so no harness can dominate it.
- Iteration 6 has the highest joint hit rate (19.3), so no harness can dominate it either.
- Iteration 8 (23.3, 12.2) is also undominated: every harness with at least its pairwise accuracy (iterations 11–14) has a lower joint hit rate (6.2–9.4).
- Every other harness is dominated. For example, the seed (6.4, 15.0) is dominated by iteration 6 (20.8, 19.3), and the final iteration 15 (21.8, 6.8) is dominated by iteration 12 (27.7, 9.4).

The frontier is therefore iterations 6, 8, and 12, the yellow bands in Figs. 3 and S6.2. It trades pairwise accuracy against joint hit rate: iteration 12 discriminates best, while iteration 6 names and locates vulnerabilities best.

Table S6.2: Validation objectives for every Inkling-Small harness (%). For dominated harnesses, one dominating harness is listed.

Iter.	Pairwise	Joint	Status
0	6.4	15.0	dominated by 6
1	11.4	13.4	dominated by 6
2	15.8	13.0	dominated by 6
3	12.9	12.5	dominated by 6
4	17.3	16.7	dominated by 6
5	14.8	14.4	dominated by 6
6	20.8	19.3	**frontier**
7	15.8	14.1	dominated by 6
8	23.3	12.2	**frontier**
9	20.8	6.8	dominated by 6
10	20.8	8.8	dominated by 6
11	24.3	9.1	dominated by 12
12	27.7	9.4	**frontier**
13	23.3	8.1	dominated by 8
14	24.8	6.2	dominated by 12
15	21.8	6.8	dominated by 12